

Machine Learning to Optimize Post-Quantum Key Generation

Randy Balzer

The Beacom College of Computer and Cyber Sciences, Dakota State University

INFA 723: Cryptography

Prof. Jason Jenkins

October 26, 2025

Machine Learning to Optimize Post-Quantum Key Generation

Abstract

Quantum computing introduces significant vulnerabilities to established cryptographic methods such as Rivest-Shamir-Adleman (RSA) and Elliptic Curve Cryptography (ECC), necessitating a transition to post-quantum cryptography (PQC) options like the lattice-based ML-KEM and code-based HQC for long term quantum resistant protection of sensitive data. However, key generation in these PQC systems demands substantial computational resources, often requiring seconds to minutes for complex tasks like lattice sampling or constructing error-correcting codes, challenges that limit their adoption in demanding environments such as high-throughput networks and Internet of Things (IoT) applications. This study proposes how machine learning techniques, including neural networks and autoencoders, can enhance PQC key generation by optimizing parameters such as lattice dimensions, polynomial degrees, or code lengths, achieving efficiency improvements of up to 25% without compromising security standards. Leveraging insights from recent research, this paper evaluates benefits such as improved integration with telecommunications frameworks, alongside concerns like potential adversarial threats to models, and examines their relevance to industry initiatives such as Nokia's Quantum Safe Networks in Malaysia. This work bridges traditional cryptographic approaches with emerging quantum-resistant solutions, presenting a pathway forward for more robust and adaptable security systems. Illustrative simulations demonstrate potential outcomes, with real experiments planned for future validation.

Keywords: post-quantum cryptography, machine learning, key generation optimization, neural networks, autoencoders, Nokia Quantum Safe Networks

Introduction

Recent advances in quantum computing, exemplified by Google's Willow chip with 105 qubits, have dramatically accelerated progress toward scalable quantum systems, enabling verifiable quantum advantage in algorithms like Quantum Echo, 13,000 times faster than supercomputers, and positioning quantum processors to break classical cryptography such as Rivest-Shamir-Adleman (RSA) and elliptic curve cryptography (ECC) through Shor's algorithm in the near term. As quantum computing rapidly

evolves, it poses profound challenges to traditional cryptographic systems, which are susceptible to quantum algorithms capable of breaking them in polynomial time (Borhan et al., 2024, as cited in Edward & Ryan, 2025). This vulnerability has accelerated the shift toward post-quantum cryptography (PQC), with the National Institute of Standards and Technology (NIST) standardizing solutions like the lattice-based ML-KEM (formerly CRYSTALS-Kyber) and the code based Hamming Quasi-Cyclic (HQC) to ensure future encryption protection (Alagic et al., 2024). However, key generation in these PQC protocols involves intensive computations, such as sampling from Gaussian distributions in lattices or decoding error-correcting codes in code-based PQC schemes like HQC, which can result in latencies of seconds to minutes, particularly on resource constrained devices like those in high throughput networks or Internet of Things (IoT) ecosystems (Al-Mohammed et al., 2024).

This paper proposes how machine learning (ML) techniques, including neural networks and autoencoders, can optimize PQC key generation by fine tuning parameters such as lattice dimensions, polynomial degrees, and code lengths. Recent studies indicate that these methods can achieve efficiency improvements of up to 25% without sacrificing security standards (Edward & Ryan, 2025). This paper addresses two primary questions: How can ML reduce computational overhead in PQC key generation while maintaining security? And what are the practical implications for real world networks, such as those developed by Nokia's Quantum Safe Networks? By focusing on ML driven efficiencies, this work contributes to the development of scalable quantum safe systems, with a particular emphasis on telecommunications applications. The urgency of transitioning to PQC is driven by quantum algorithms like Shor's and Grover's that threaten classical systems, a concern being addressed by ongoing NIST standardization efforts (Alagic et al., 2024). As quantum computing advances, efficient PQC implementations become increasingly critical, particularly on hardware constrained platforms (Yi, 2023). The Experimental Results section presents illustrative simulations to demonstrate potential outcomes, with real experiments planned for future validation.

Background and Literature Review

Post-Quantum Cryptography Fundamentals

Post-quantum cryptography emerged as a response to quantum threats, with lattice-based schemes like ML-KEM relying on the hardness of learning with errors (LWE) problems and code-based

schemes like HQC depending on the difficulty of decoding random codes (Chhetri et al., 2025). Key generation in ML-KEM involves matrix operations and polynomial sampling, while HQC focuses on constructing quasi-cyclic codes, both techniques of which demand significant computational resources (Alagic et al., 2024). These key generation processes can exceed seconds on edge devices, posing barriers to widespread adoption in IoT and high throughput environments (Al-Mohammed et al., 2024). The computational complexity in key generation arises from the need to ensure security against quantum attacks, a challenge that NIST's standardization process continues to address through iterative rounds (Alagic et al., 2024). Lattice-based schemes leverage the mathematical difficulty of problems like LWE, while code based approaches utilize decoding challenges, both implementations requiring optimization for practical deployment, especially in hardware contexts (Yi, 2023).

Machine Learning in Cryptography

The intersection between machine learning and cryptography is rapidly increasing, particularly regarding techniques for optimization. Neural networks excel in predicting patterns in data, enabling forecasting of optimal parameters for key generation (Edward & Ryan, 2025). For example, supervised models trained on simulation data can identify optimized lattice dimensions to minimize computational load, particularly on low power hardware platforms (Yi, 2023). Autoencoders are able to predict key related metrics with high accuracy in scalable systems (Al-Mohammed et al., 2024). These techniques are particularly valuable in adapting PQC protocols to scalable constraints (Edward & Ryan, 2025). ML enables automated key management, providing strategies for efficient key distribution and rotation (Edward & Ryan, 2025).

Gaps in Existing Research

Recent studies show that ML can reduce PQC operation times by up to 25% (Edward & Ryan, 2025), however empirical research on key generation optimization remains limited, particularly regarding hybrid threats like adversarial ML attacks (Chhetri et al., 2025). This paper addresses these gaps by proposing ML for parameter optimization, building on the latest advancements in the field (Edward & Ryan, 2025). The lack of comprehensive datasets for training ML models in high-throughput key processes remains a critical area for further exploration (Al-Mohammed et al., 2024). Additionally, the

integration of ML with PQC in real-world hardware applications requires more empirical validation (Yi, 2023).

Methodology

Proposed Dataset and Model Design

If real experiments were conducted, this study would utilize simulated datasets from PQC key generation runs, including metrics such as computation time, resource usage, and security levels for ML-KEM and HQC. Data would be generated using libraries like OpenQuantumSafe, comprising 10,000 samples with features, such as lattice dimensions, and targets, such as optimal configurations. Neural networks would form the core model, employing a feedforward architecture with ReLU activations to predict cryptographic parameters from side channel data (Yi, 2023). Autoencoders would predict key related metrics with high accuracy (Al-Mohammed et al., 2024). The dataset would include diverse scenarios, ranging from high performance servers to low power IoT devices, to ensure generalizability across PQC implementations (Chhetri et al., 2025). Data preprocessing would involve normalizing features to enhance model convergence.

Proposed Integration with PQC Protocols

Optimization would target ML-KEM's polynomial sampling and HQC's code construction. Models would input current system constraints, such as memory, processing power, and output adjusted tuned parameters, integrated via model API calls to PQC implementations.

Figure 1. High-Level Pseudocode for Neural Network Optimization of PQC Key Generation

```
# Import conceptual libraries (not executable)
import NeuralNetworkLibrary as nn
import DataProcessing as dp

# Step 1: Prepare Dataset
historical_data = load_dataset("pqc_key_gen_history.csv")
features = dp.extract_features(historical_data, columns=["comp_time",
"resource_usage", "security_level"])
targets = dp.extract_targets(historical_data, columns=["optimal_lattice_dim",
"optimal_poly_size"])

# Step 2: Define Neural Network Model
model = nn.SequentialModel()
model.add_layer(nn.DenseLayer(input_size=features.shape[1],
output_size=64, activation="relu"))
model.add_layer(nn.DenseLayer(input_size=64, output_size=32, activation="relu"))
model.add_layer(nn.DenseLayer(input_size=32, output_size=2, activation="linear"))

# Step 3: Train the Model
optimizer = nn.AdamOptimizer(learning_rate=0.001)
loss_function = nn.MeanSquaredError()
model.train(features, targets, epochs=100, batch_size=32,
optimizer=optimizer, loss=loss_function)

# Step 4: Predict Optimized Parameters
new_input = dp.prepare_input([new_comp_time, new_resource_usage, new_security_level])
optimized_params = model.predict(new_input)

# Step 5: Integrate with PQC Key Generation
pqc_key_gen = initialize_pqc_algorithm("ML-KEM", lattice_dim=optimized_params[0],
poly_size=optimized_params[1])
generated_key = pqc_key_gen.execute()
```

The integration process would involve real-time parameter adjustments, ensuring compatibility with existing PQC libraries and supporting dynamic adaptation to hardware constraints. This approach would allow for tailored optimizations based on specific hardware and security requirements (Edward & Ryan, 2025).

Evaluation Metrics

Efficiency would be measured by time reductions, targeting up to 25%, with additional metrics such as energy consumption and memory usage evaluated to evaluate performance on IoT devices. Security would be assessed via indistinguishability tests and robustness against attacks (Chhetri et al., 2025), including hardware specific optimizations (Yi, 2023). These metrics would provide a comprehensive view of optimization impact across diverse environments (Al-Mohammed et al., 2024). Security evaluations would include testing against simulated quantum attacks to ensure resilience (Alagic et al., 2024). Tools used would include TensorFlow for ML and liboqs for PQC.

Experimental Results

For illustration, simulations on a standard CPU (Intel i7, 16GB RAM) or equivalent cloud resources tested models on 1,000 validation samples. Neural networks achieved a hypothetical 22% average time reduction in ML-KEM key generation, from 450 ms to 351 ms, with hardware optimizations enhancing performance. Autoencoders yielded hypothetical 18% gains in dynamic scenarios. Table 1 summarizes the results:

Table 1. Hypothetical Efficiency Gains in PQC Key Generation

Algorithm	Baseline Time (ms)	Optimized Time (ms)	Gain (%)
ML-KEM (Neural)	450	351	22
HQC (Autoencoder)	620	508	508

Note. Hypothetical data from 1,000 simulations. Security levels maintained per NIST standards (Alagic et al., 2024).

These illustrative simulations suggest robustness across varying network conditions, with hypothetical resource savings of up to 25% in IoT scenarios by leveraging hardware efficiencies (Edward & Ryan,

2025). These hypothetical results illustrate the potential scalability of ML optimizations, with consistent performance across different PQC implementations (Edward & Ryan, 2025). The simulations also highlights the importance of dataset diversity in achieving reliable outcomes (Al-Mohammed et al., 2024).

Discussion

The illustrative experiment results suggest ML's role in reducing PQC overhead is impactful, with up to 25% gains aligning with projections (Edward & Ryan, 2025). The hypothetical 22% reduction in ML-KEM key generation time demonstrates neural networks' effectiveness in predicting optimal lattice parameters, while the 18% gain for HQC highlights autoencoders' applicability in code-based scenarios. These results underscore ML's potential to reduce computational bottlenecks in resource constrained environments, such as IoT and high-throughput 5G networks, where Nokia's Quantum Safe Networks (QSN) deployment in Malaysia highlights practical PQC integration for low latency enterprise security (Maxis & Nokia, 2024). Nokia's QSN leverages PQC algorithms like ML-KEM to secure 5G backhaul and enterprise VPNs, addressing quantum threats in high bandwidth telecom while maintaining NIST recommended security levels (Maxis & Nokia, 2024; Alagic et al., 2024). In this context, ML optimized key generation could enable dynamic parameter adjustment for Nokia's QSN, reducing setup latency from milliseconds to microseconds in edge deployments, extremely significant for real-time applications like video streaming and remote surgery. For instance, neural networks could predict polynomial degrees for ML-KEM based on network load, ensuring efficiency without compromising IND-CCA2 security, thus supporting Nokia's goal of quantum-safe, scalable infrastructure. However, integrating ML into QSN introduces challenges like adversarial robustness in dynamic 5G channels, where radiation or interference could compromise models. Limitations include data dependency and risks of adversarial attacks, where models could be compromised (Chhetri et al., 2025).

Beyond efficiency, ML can bolster PQC security against adversarial examples in key generation. For instance, neural networks trained on perturbed inputs could detect and mitigate poisoning attacks on parameter predictions, ensuring reliable lattice sampling in ML-KEM even under noisy conditions (Chhetri et al., 2025). In Nokia QSN contexts, this robustness is vital for 5G edges where environmental factors might skew ML outputs. Simulations could test GAN-based adversaries to validate defenses,

aligning with the proposed methodology's emphasis on secure optimization. This dual focus on speed and resilience underscores ML's holistic value in quantum-safe systems. However, integrating ML into QSN introduces challenges like adversarial robustness in dynamic 5G channels, where radiation or interference could compromise models. Limitations include data dependency and risks of adversarial attacks, where models could be compromised (Chhetri et al., 2025). Future work might explore advanced ML integrations to further enhance performance, including real experiments to validate these simulations in Nokia like environments.

Challenges and Limitations

While the initial results of ML in optimizing post-quantum key generation are encouraging, several challenges exist. First, ML models require large datasets for training, which are often scarce in cryptographic contexts due to security constraints (Chhetri et al., 2025). Additionally, the computational overhead of training ML models can offset efficiency gains, particularly for resource constrained devices, such as IoT systems (Al-Mohammed et al., 2024). Also, ensuring the security of ML optimized key generation against adversarial attacks is critical, as models may introduce vulnerabilities (Chhetri et al., 2025). Finally, hardware specific challenges, such as side-channel vulnerabilities, further hinder deployment (Yi, 2023). These challenges demonstrate the need for ongoing research into lightweight and secure ML frameworks (Chhetri et al., 2025).

Future Directions

The integration of ML into post-quantum key generation offers several promising directions. Firstly, by developing lightweight ML models for resource constrained environments, such as IoT devices, could broaden their applicability (Chhetri et al., 2025). In addition, hybrid approaches combining ML with traditional methods for encryption key generation may balance efficiency and security (Edward & Ryan, 2025). Thirdly, advances in hardware acceleration could further reduce latency in real-world deployments. Also, collaboration between the machine learning and cryptography communities are essential to establish benchmarks and datasets for robust ML models. Finally, industry partnerships could accelerate deployment in real-world networks, particularly in high throughput environments. Future research should prioritize standardized ML benchmarks and integration with emerging quantum hardware. Real experiments to validate the proposed methodology are planned.

Conclusion and Future Work

This paper proposes how ML bridges classical and PQC encryption for efficient key generation, supporting scalable security. This survey highlights the transformative potential of machine learning in optimizing post-quantum key generation. By examining ML applications in lattice-based, code-based, and multivariate polynomial-based schemes, we highlight how techniques like neural networks and autoencoders enhance efficiency and security. Challenges such as data scarcity and model security must be addressed (Chhetri et al., 2025). One area of future research could include the investigation of advanced ML integrations for stronger efficiency gains (Edward & Ryan, 2025). An additional area of research could focus on lightweight ML models and hybrid approaches for encryption key parameter tuning in PQC key generation. As quantum computing quickly evolves, the synergy between ML and PQC will be crucial for developing secure and efficient cryptographic systems. Real experiments to validate the illustrative simulations are planned.

References

- Alagic, G., Alperin-Sheriff, J., Apon, D., Cooper, D., Dang, Q., Kelsey, J., Liu, Y.-K., Miller, C., Moody, D., Peralta, R., Perlner, R., Robinson, A., & Smith-Tone, D. (2022). *Status report on the third round of the NIST post-quantum cryptography standardization process* (NIST Interagency or Internal Report No. 8413-upd1). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.IR.8413-upd1>
- Al-Mohammed, H. A., Al-Kuwari, S., Kuniyil, H., & Farouk, A. (2024). *Towards scalable quantum key distribution: A machine learning-based cascade protocol approach*. arXiv. <https://doi.org/10.48550/arXiv.2409.08038>
- Chhetri, G., Somvanshi, S., Hebli, P., Brotee, S., & Das, S. (2025). *Post-quantum cryptography and quantum-safe security: A comprehensive survey*. arXiv. <https://doi.org/10.48550/arXiv.2510.10436>
- Edward, E., & Ryan, G. (2025). *AI-driven post-quantum cryptographic key management techniques*. *Revista de Inteligencia Artificial en Medicina*, 16(1), 90–103. <https://www.researchgate.net/publication/391021627>
- Yi, H. (2023). *Machine learning method with applications in hardware security of post-quantum cryptography*. *Journal of Grid Computing*, 21(1), Article 19. <https://doi.org/10.1007/s10723-023-09643-4>